

7

Regression and Correlation Analysis

7
(1)

Simple Linear

Regression Analysis

7.1 Simple Linear Regression Analysis

Suppose you are planning to sell your house and need to set a selling price.

To determine a fair and reasonable price, what factors should you consider?

7.1 Simple Linear Regression Analysis

To arrive at a fair selling price, you must consider:

- ❑ what you paid for the house,
- ❑ what has been happening to the price of houses since you bought yours,
- ❑ the actual selling price of houses in your neighborhood,

7.1 Simple Linear Regression Analysis

- ❑ the size of houses in your neighborhood,
- ❑ the size of your house,
- ❑ the location of your house and many other variables.

7.1 Simple Linear Regression Analysis

Such kind of information proves useful in predicting how you expect the buyers are likely to react to your house.

Here, what we need is finding a suitable mathematical relationship between the one unknown variable called the *dependent variable* and the group of other known quantities called *independent variables*.

7.1 Simple Linear Regression Analysis

- This example involves a complex relationship between one unknown variable (the selling price of a house) and a collection of other variables.

One methodology for handling this type of problems is called *regression analysis*.

7.1 Simple Linear Regression Analysis

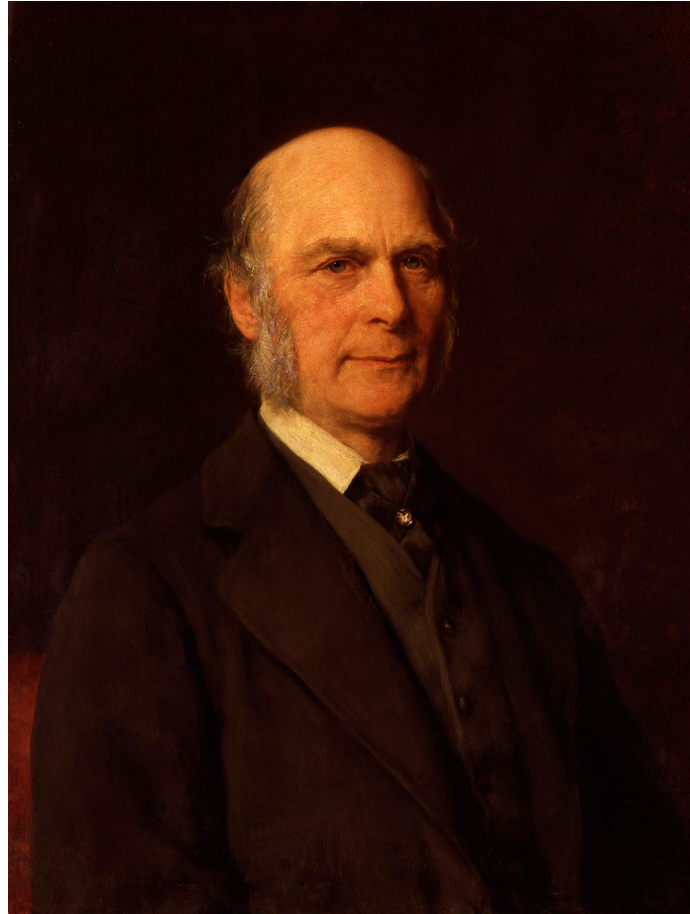
Definition 1:- A quantitative expression of the basic nature of the relationship between a *dependent variable* and the values of one or more *independent variables* on which it depends is called *regression*.

7.1 Simple Linear Regression Analysis

Definition2:- *Regression analysis* is a mathematical measure of the average relationship between two or more variables in terms of the original units of data.

☐ Regression was first used by the British biometrician **Sir Francis Galton** (1822 -1911).

7.1 Simple Linear Regression Analysis



7.1 Simple Linear Regression Analysis

Regression is concerned with the “prediction” of the most likely value of one variable when the value of the other variable is known.

Thus, the term regression is used to denote *estimation or prediction* of the average value of one variable for a specified value of the other variable.

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□ The estimation is done by means of suitable equation, derived on the basis of available data.

□ In regression analysis, there are two types variables and they are:

✓ *independent variable(s)* and

✓ *dependent variable*

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In regression, we have only *one dependent variable*. However, we can have *more than one independent variable*.

The situation where we have only one independent variable is called *simple regression*.

If the mathematical model relating the dependent variable to the independent variable is linear, then the relationship is referred to as *simple linear regression*.

7.1 Simple Linear Regression Analysis

Unlike linear regression, which fits a straight line, non-linear regression fits a curve to the data using numerical optimization to find the best parameters for a given non-linear function, often derived from theoretical principles.

7.1 Simple Linear Regression Analysis

Non-linear regression is used to model data with complex, non-linear relationships between variables.

For example:

1. Researchers use non-linear regression to model drug concentration in the bloodstream over time, accounting for how drugs are absorbed, metabolized, and excreted.

7.1 Simple Linear Regression Analysis

2. Predicting population growth, which often follows a non-linear pattern over time, can be done using non-linear regression.
3. Modeling the relationship between economic indicators like gold prices and inflation, or investor sentiment and stock market returns, can involve non-linear regression to capture complex trends.

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4. Predicting the probability of patient recovery from an illness can be done with a logistic regression model, which is a type of non-linear regression used for binary outcomes.

Non-linear regression can model more complex relationships between variables than linear regression can, which is limited to straight-line fits.

7.1 Simple Linear Regression Analysis

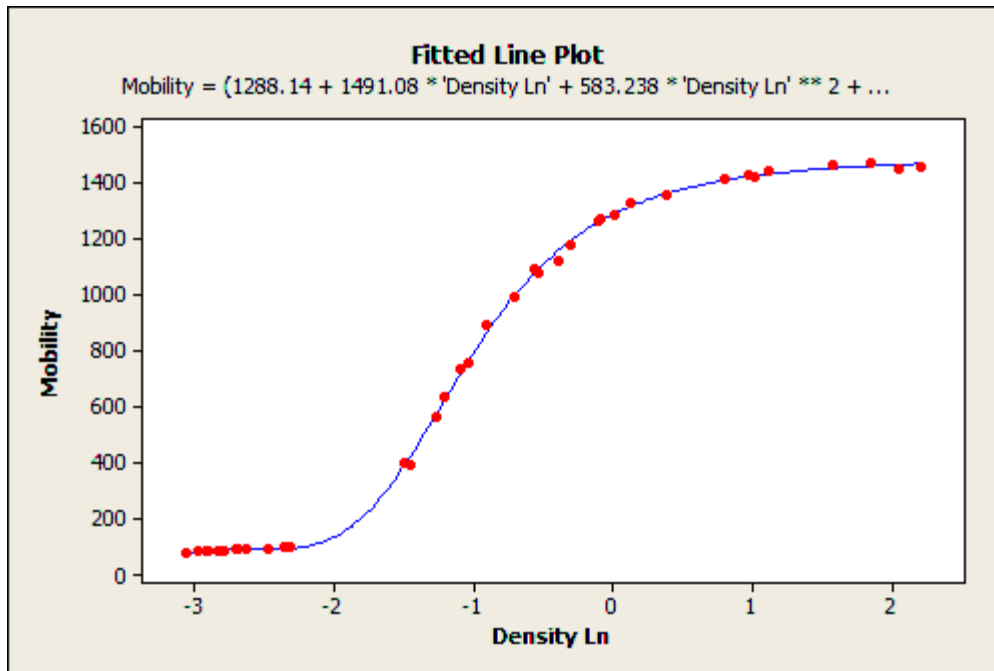
If regression is not linear, it is known as non-linear.

Some types of non-linear regressions are:

1. Parametric non-linear regression
2. Non-parametric non-linear regression
3. Polynomial non-linear regression

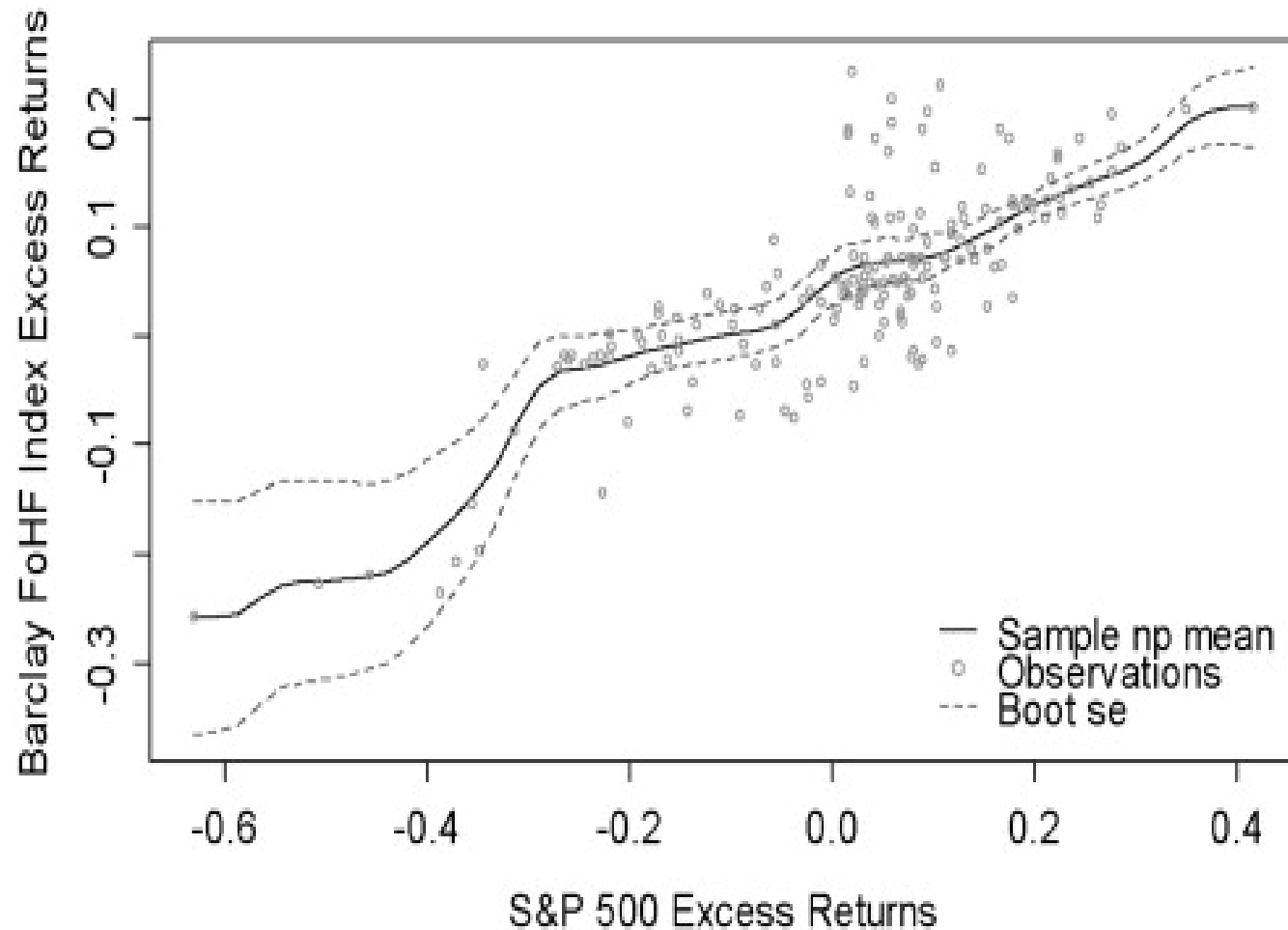
$$y = \beta_0 + \beta_1 X + \beta_2 X^2 + . . . + \beta_n X^n$$

7.1 Simple Linear Regression Analysis

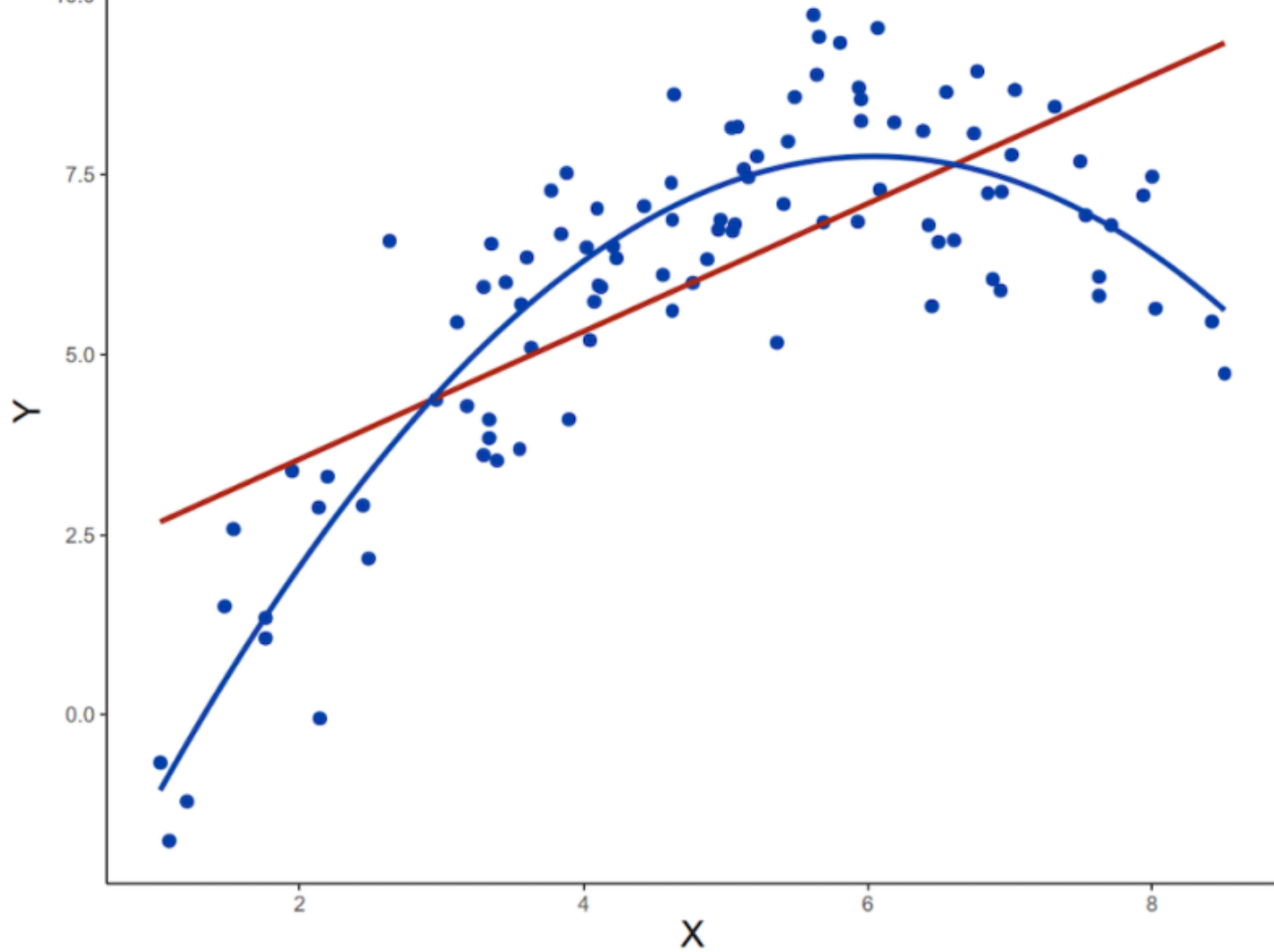


Parametric non-linear regression

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Non-parametric non-linear regression



Polynomial non-linear regression

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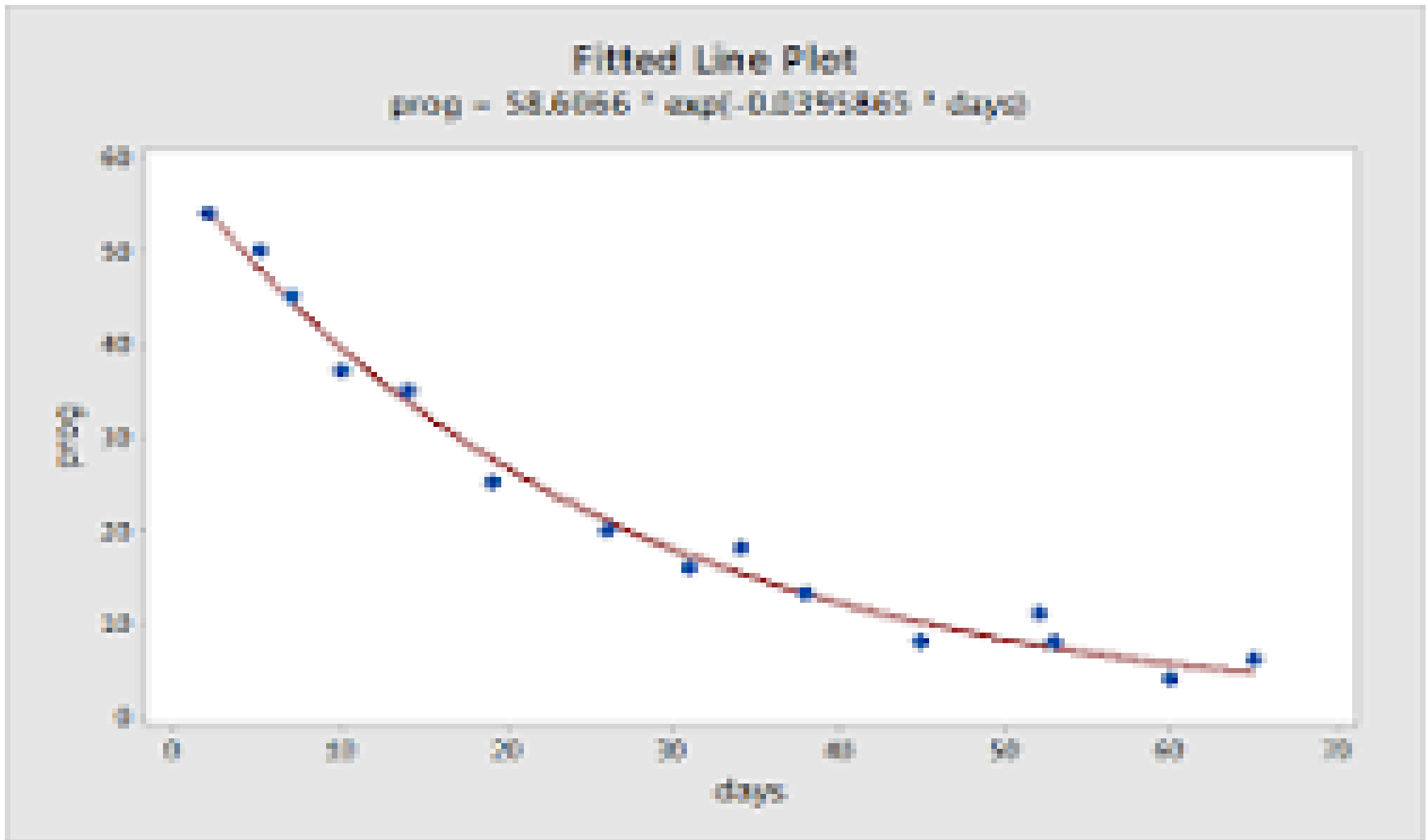
4. Exponential non-linear regression:

$$y = \alpha e^{\beta x}$$

5. Logarithmic non-linear regression:

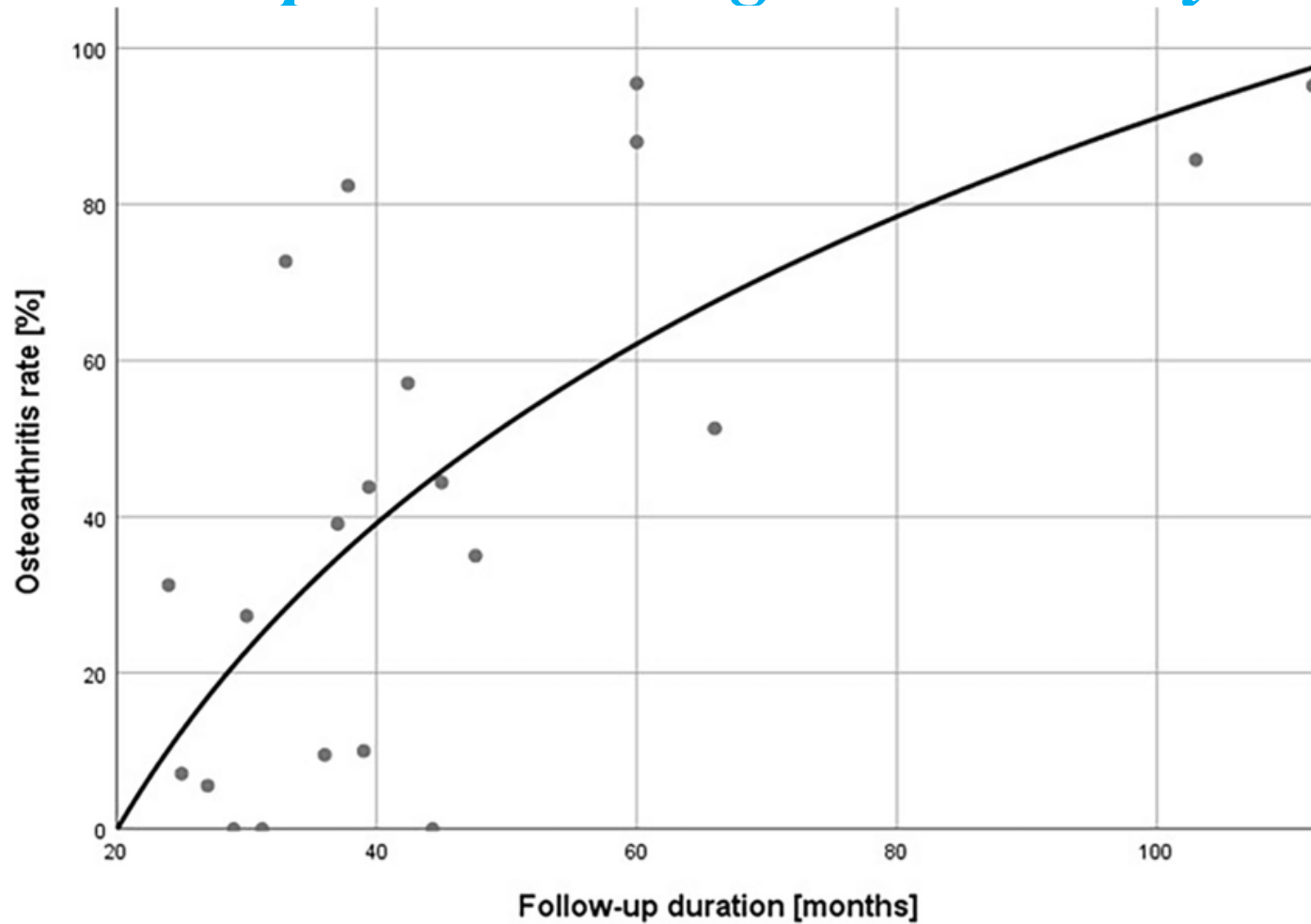
$$y = \alpha + \beta \ln x$$

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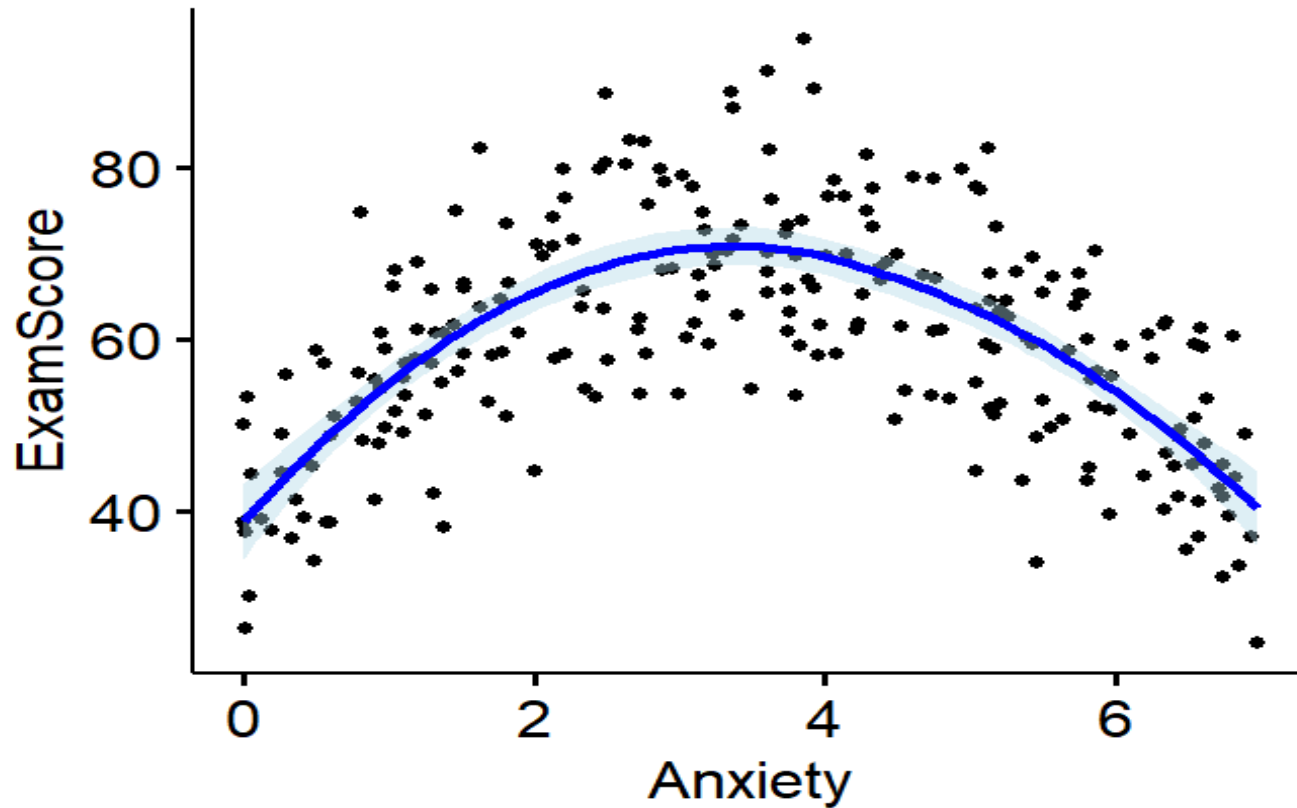
Exponential non-linear regression

7.1 Simple Linear Regression Analysis



Logarithmic non-linear regression:

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Exam scores vs Anxiety

7.1 Simple Linear Regression Analysis

You can use multiple linear regression when you want to know: How strong the relationship is between two or more independent variables and one dependent variable (e.g. how rainfall, temperature, and amount of fertilizer added affect crop growth).

7.1 Simple Linear Regression Analysis

Scenario: Departure delay (minutes) is modeled by flight distance (in 100s of km), rainfall at departure (mm), and wind speed (knots):

$$\hat{\text{Delay}} = 3.2(\text{Distance}) + 0.40(\text{Rain}) + 0.20(\text{Wind}) + 5$$

Predict the delay for Distance = 15 (i.e., 1500 km),
Rain = 6 mm, Wind = 12 knots.

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A few years ago, I started using a case study from China: a company wanted to be able to estimate or predict how much fuel they needed to transport building materials to their oil wells so that they could line them with concrete.

Duncan Williamson

7.1 Simple Linear Regression Analysis

The data provided was:

- Number of wells
- Depth of wells
- Distance to wells
- Weight of materials
- Fuel costs

7.1 Simple Linear Regression Analysis

- The full multiple regression model from these data is

$$Y = a + b_1 * \text{No Wells} + b_2 * \text{Depth} \\ + b_3 * \text{Distance} + b_4 * \text{Weight}$$

Where Y is the Fuel Cost Prediction

7.1 Simple Linear Regression Analysis

In this chapter we are going to study the mathematical model which is relating the dependent variable to one independent variable. That is we only will consider a *simple linear regression*.

7.1 Simple Linear Regression Analysis

Example 1:- Suppose the relationship between the average daily expenditure and income (both in birr) of households is expressed by:

$$\text{Daily average expenditure} = 0.6(\text{daily income}) + 30$$

Here, on the basis of daily income, we can predict average daily expenditure.

7.1 Simple Linear Regression Analysis

For instance, if the daily income of a certain household is 225 birr, then the estimated average daily expenditure will be:

$$\text{Daily average expenditure} = 0.6(225) + 30$$

$$\text{Daily average expenditure} = 135 + 30 = 165 \text{ br.}$$

7.1 Simple Linear Regression Analysis

Note that since the average daily expenditure is estimated on the basis of daily income, *expenditure is the dependent* variable and *income is the independent* variable.

Dependent variable (y): The variable whose value is influenced or is to be predicted is called dependent variable.

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- The regression equation of y on x describes the changes in the value of y for given changes in the value of x .
- A line fitted to a set of data points to estimate the relationship between the two variables is called the *regression line* or *the estimated regression line* or *the least square line*.

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Simple regression line is a straight line that describes the dependence of the average value of one variable on the other.

The equation is given by $y = \beta_0 + \beta_1 x$,

where y – dependent or outcome variable

x – independent or predictor variable

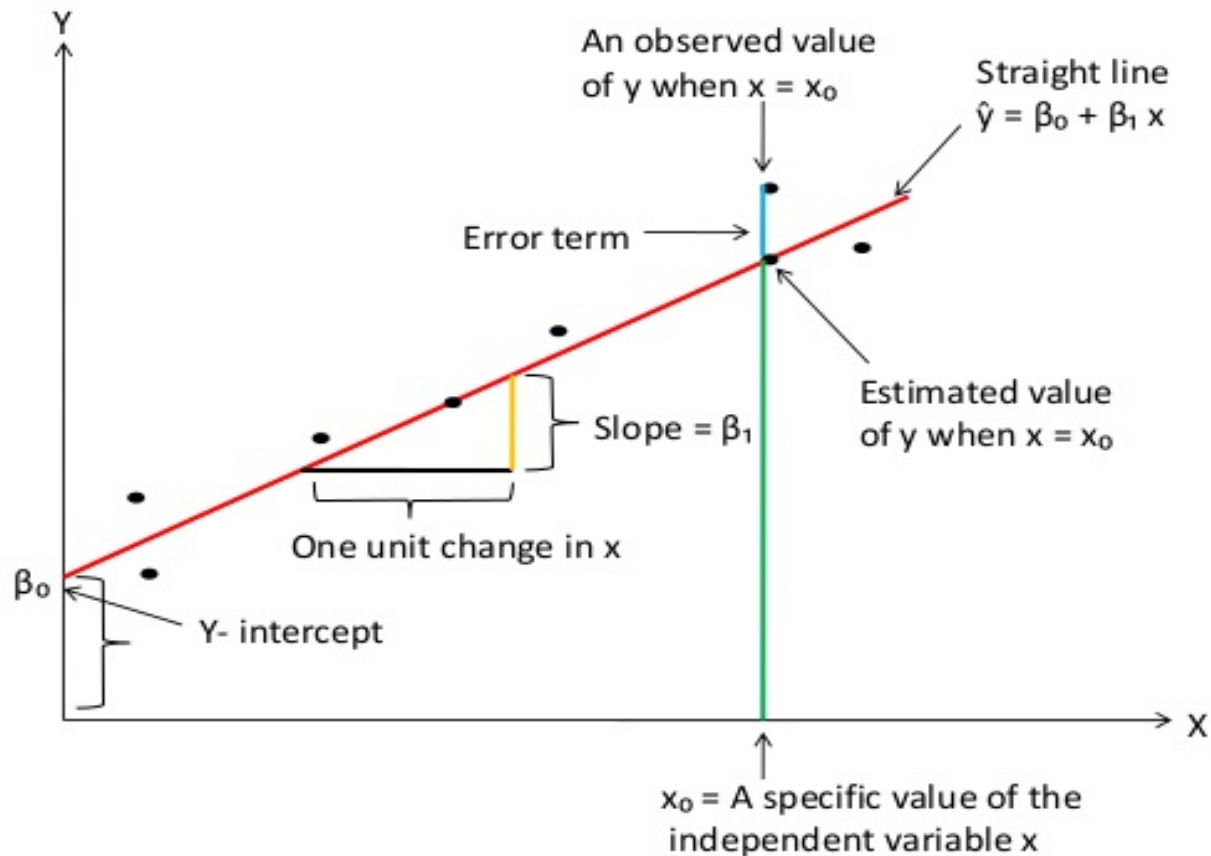
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β_0 – y-intercept of the model

β_1 – Population slope(or gradient) coefficient.

It is also known as population regression coefficient. It measures the average rate of change in dependent variable per unit change in independent variable.

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7.1 Simple Linear Regression Analysis

Since not all the points of the given data lie on the regression line, we introduce the *estimated equation of the regression line* by:

$$\hat{y} = b_0 + b_1x,$$

where $\hat{y}$ - the estimated regression line

b_0 - the y – intercept of the model

b_1 - the slope (or gradient) coefficient

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The computational formula to find b_1 and b_0 respectively are given as follows:

$$b_1 = \frac{n\sum xy - \sum x \sum y}{n\sum x^2 - (\sum x)^2} \quad \text{and} \quad b_0 = \bar{y} - b_1 \bar{x}$$

$$\text{where } \bar{y} = \frac{\sum y}{n} \quad \text{and} \quad \bar{x} = \frac{\sum x}{n}$$

7.1 Simple Linear Regression Analysis

Example 2:- The following table shows the number of items produced and the cost incurred in producing them (in birr).

Number of items produced	Cost incurred (in birr)
4	15
5	18
6	18
8	20
9	22

7.1 Simple Linear Regression Analysis

- Find the equation of the regression line
- Identify the slope and the y-intercept and interpret them.
- Estimate the cost of producing 7 items.

Number of items to be produced (x)	Cost incurred (y)	xy	x^2
4	15	60	16
5	18	90	25
6	18	108	36
8	20	160	64
9	22	198	81
$\sum_1^5 x_i = 32$	$\sum_1^5 y_i = 93$	$\sum_1^5 x_i y_i = 616$	$\sum_1^5 x_i^2 = 222$

7.1 Simple Linear Regression Analysis

$$b_1 = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2} = \frac{5(616) - (32)(93)}{5(222) - (32)^2}$$
$$= \frac{3080 - 2976}{1110 - 1024} = \frac{104}{86} = 1.21$$

$$\bar{y} = \frac{\sum y}{n} = \frac{93}{5} = 18.6 \quad \bar{x} = \frac{\sum x}{n} = \frac{32}{5} = 6.4$$

$$b_0 = \bar{y} - b_1 \bar{x} = 18.6 - 1.21(6.4) = 18.6 - 7.74 = 10.86$$

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- Thus the estimated equation of the regression line is given by

$$\hat{y} = b_0 + b_1x \Rightarrow \hat{y} = 10.86 + 1.21x$$

- The slope (gradient) is **1.21** and this number indicates that as the number of items to be produced is increased by 1, the cost incurred to produce the items averagely increases by 1.21 birr.

7.1 Simple Linear Regression Analysis

- The y-intercept is **10.86**. It can be found by making $x = 0$. It indicates that even if there is no item to be produced, there will be an average cost of 10.86 birr which might be electricity and water bills, insurance, and so on.

7.1 Simple Linear Regression Analysis

Example 2:- The Quality Assurance and Student Service of HiLCoE wants to determine whether there is any type of relationship between the amount of alumni's contribution (in tens of birr) and the years the alumnus has been out of schools. The following table refers to the contribution of the sample of five students and the number of years they graduated from HiLCoE.

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Number of years	Contribution (in tens of birr)			
1	50			
5	10			
7	6			
3	30			
6	8			

7.1 Simple Linear Regression Analysis

- i. Find the estimated equation of the regression line.
- ii. Identify the slope (gradient and interpret your result.
- iii. Estimate the contribution of an alumna if the number of years of graduation is four years.

Any Questions



Any Questions

